Please attach this cover sheet to your completed homework

Homework 1 (Ch. 7) **ECE 311 – Hang Gao**

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Problem	Course outcome	Grade
1	1.c.	/20
2	1.c.	/20
3	1.c.	/20
4	1.c.	/20
5	1.c.	/10
6	1.c.	/10
	TOTAL:	/100

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #1
Due date: Sunday, Feb. 1st by 11:59 PM (Electronically – Canvas)

1. (20 points) The voltage and current of an electric load can be expressed by the following equations:

$v = 340\sin(628.318t + 0.5236)$ V

$i = 100\sin(628.318t + 0.87266)$ V

Note that the angles are in radians.
Calculate the following:

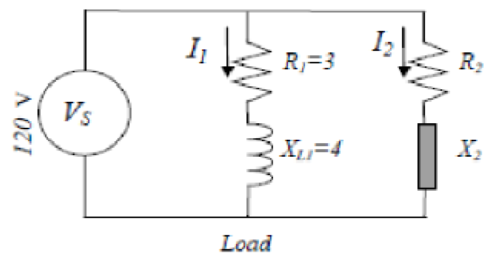
a. The rms voltage
b. The frequency of the current
c. The phase shift between current and voltage in degrees
d. The average voltage
e. The load impedance

2. (20 points) An electric load consists of a 4 Ω resistance, a 6 Ω inductive reactance, and an 8 Ω capacitive reactance connected in series. The total impedance of the load is connected across a voltage source of 120V:
a. Compute the power factor of the load
b. Compute the source current
c. Compute the real power of the circuit
d. Compute the reactive power of the circuit

3. (20 points) A voltage source $\overline{V} = 120\angle 30°$ is connected across a circuit consisting of R=3.5 Ω in series with L = 0.0833 H. If the circuit is energized in the united states (the power supply frequency is 60Hz), compute the following:
a. The load current
b. The load power factor
c. The real power consumed by the load
d. The reactive power consumed by the load

4. (20 points) A voltage source $\overline{V} = 120\angle 30°$ is connected to a circuit consisting of R=3.5 Ω in series with L = 0.0833 H. If the circuit is energized in Europe (the power supply frequency is 50Hz), compute the following:
a. The load current
b. The load power factor
c. The real power consumed by the load
d. The reactive power consumed by the load

5. (10 points) An inductive load consists of R_1 = 3 Ω in series with X_{L1} = 4 Ω. The load is connected in parallel with another load of unknown impedance (R_2+jX_2). The voltage source of the system is 120V. The total real and reactive powers delivered by the source are 3 kW and 2 kVAR, respectively. Compute the impedance of the unknown load.



6. (10 points) An inductive load consists of 1 Ω resistance and 1.0mH inductance connected in series. The load is connected across 120V, 60Hz source.

a. Compute the power factor of the load.

b. A capacitor is connected in parallel with the entire load so that no reactive power is delivered by the source. Compute the value of the capacitance.